Are Close Binaries Formed Through Disk Fragmentation?

2017.1.00053.S

ABSTRACT

Binary and multiple star formation is a common outcome of the star formation process, likely occuring at early times. To characterize the multiplicity frequency versus separation, we have conducted a VLA 8mm survey of all protostars (N~80) in the Perseus molecular cloud (d=230pc) at 0.065" (15 AU) resolution, identifying 18 proto-binary systems with separations between 18.5 and 440 AU. However, the VLA data alone do not directly constrain the formation scenarios of these close multiples, which are likely either disk fragmentation or turbulent fragmentation with dynamical evolution. We can discern between disk and turbulent fragmentation by characterizing the environments around the multiple systems. Therefore, we propose for ALMA observations of dust continuum and molecular line emission in C18O, 13CO, and 12CO toward 12 of the 18 close multiples. With these observations, we will identify rotationally-supported circumbinary disks, aligned or misaligned outflows, and relative source velocities that will reveal the origins of these systems. This sample is large enough to reveal whether or not close multiples typically form via disk fragmentation or turbulent fragmentation.

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ESTIMATED 12M TIME:	**5.9 h**	**ESTIMATED ACA TIME:**	**0.0 h**	**ESTIMATED NON-STANDARD MODE TIME (12-M):**	**0.0 h**

CO-PI NAME(S): (Large & VLBI Proposals only)	
CO-INVESTIGATOR NAME(S):	Leslie Looney; Dominique Segura-Cox; Erin Cox; Robert Harris; Zhi-Yun Li; Kaitlin Kratter; Claire Chandler; Laura Perez; Carl Melis; Michael Dunham; Patrick Sheehan; Nickalas Reynolds; Brian Stephenson; Sarah Sadavoy
DUPLICATE OBSERVATION JUSTIFICATION:	These observations will be repeating 12/18 sources from the partially completed program 2013.1.00031.S. That proposal was only 25% completed and did not meet sensitivity requirements. The poor spectral line sensitivity of that proposal requires reobservation to achive high enough signal-to-noise achieve our science goals

REPRESENTATIVE SCIENCE GOALS (UP TO FIRST 30)

SCIENCE GOAL	POSITION	BAND	ANG.RES.(")	LAS.(")	ACA?	NON-STANDARD MODE
Perseus Binaries	ICRS 03:29:10.5360, 31:13:30.930	6	0.250	3.500	N	N
Total # Science Goals : 1						

SCHEDULING TIME CONSTRAINTS	NONE	**TIME ESTIMATES OVERRIDDEN ?**	No

Proposal History: This is a resubmission of the B-rated/25% completed Cycle 2 program 2013.1.00031.S, the C-rated Cycle 3 program 2015.1.00316.S (no observations), and the B-rated Cycle 4 program 2016.1.00058.S (no observations thus far).

1 Scientific Justification

Binary and multiple star systems are frequent products of the star formation process; over 50% of solar type field stars are found in binaries or higher order multiple systems, with an average separation of ∼50 AU (Ragahvan et al. 2010, Lada 2006). Thus, key questions in the star formation process include (1) how multiples themselves form and (2) what factors determine their separations. Multiple formation is expected to begin during the earliest phases of star formation, the Class 0 phase (André et al. 1993), when the largest mass reservoir is available (e.g. Tohline 2002). Thus, to get at the heart of multiple star formation, the youngest protostars must be examined. However, the dense enshrouding envelopes make multiplicity difficult to characterize in the Class 0 phase, thus (sub)millimeter interferometry is the ideal method to characterize these systems.

The formation of mechanism of close multiple protostar systems at separations <500 AU is an area of significant debate. Two primary routes may lead to close multiple formation, either disk fragmentation via gravitational instability (e.g. Kratter et al. 2010; Vorobyov 2010) or turbulent/rotational fragmentation at initially larger separations (1000 AU) with inward migration (Offner et al. 2010). Earlier attempts to characterize multiplicity toward Class 0 sources have offered conflicting results as to the frequency and distribution of companion separations and likely formation mechanisms (Looney et al. 2000; Maury et al. 2010; Chen et al. 2013). These pioneering efforts were all hampered by small and biased samples, inhomogeneous sensitivity, and variable resolution.

We have recently completed the VLA Nascent Disk and Multiplicity Survey (VANDAM) toward all known protostars (N ∼80) in the Perseus molecular cloud with the VLA at 8 mm with a spatial resolution of 15 AU toward all sources (Tobin et al. 2016). This survey has substantially improved our knowledge of protostellar multiplicity, finding that ∼60% of Class 0 systems are in multiples (consistent with Chen et al. 2013) and that companions at separations <500 AU comprise ∼50% of all multiples. We discovered 18 multiple protostar systems with separations < 500 AU, 15 of which are new discoveries and have with a median separation is ∼75 AU. This sample of close multiples has roughly equal numbers of Class 0 and Class I sources, with bolometric luminosities ranging from ∼1 $L_\odot$ to 30 $L_\odot$. This is the most complete sample of close multiples discovered with a uniform survey of a single star-forming region, and a subsample (12 of 18) is shown in Figure 1.

The VLA continuum data, however, only reveal the presence of the multiple system and very little about the circumbinary environment or formation mechanism. To make further progress, the dust continuum and the gas kinematics of the circumbinary environment must be characterized to ascertain the most likely formation mechanism for these close multiple systems.

2 Proposed Observations

We propose ALMA observations of 12 close multiple systems at ∼0.25″ resolution (60 AU) in 1.3 mm continuum and the $C^{18}O$, ^{13}CO, CO ($J = 2 \rightarrow 1$) molecular lines to characterize their circumbinary environments. The results from our 25% completed ALMA Cycle 2 program indicate that these 12 are the most promising for detecting gas kinematics and determining the most likely formation mechanism. This conclusion based on their continuum emission and weak line detections in the current data. The resolution of the observations will be 0.35″ x 0.2″ (80 AU x 46 AU), enabling most individual sources to be resolved and recover extended structure out to ∼3.5″(800 AU). The continuum will reveal the cold dust emission around these systems, providing dust mass estimates for the circumbinary environment and most individual components.

The molecular line data will reveal the kinematics of the dense gas surrounding the sources ($C^{18}O$ and ^{13}CO) to determine if the companions are embedded within a rotating disk structure and if that rotation is Keplerian. We will also be able to look for outflow emission on small-scales (CO) to determine if each source is driving a detectable outflow and the relative alignment between them.

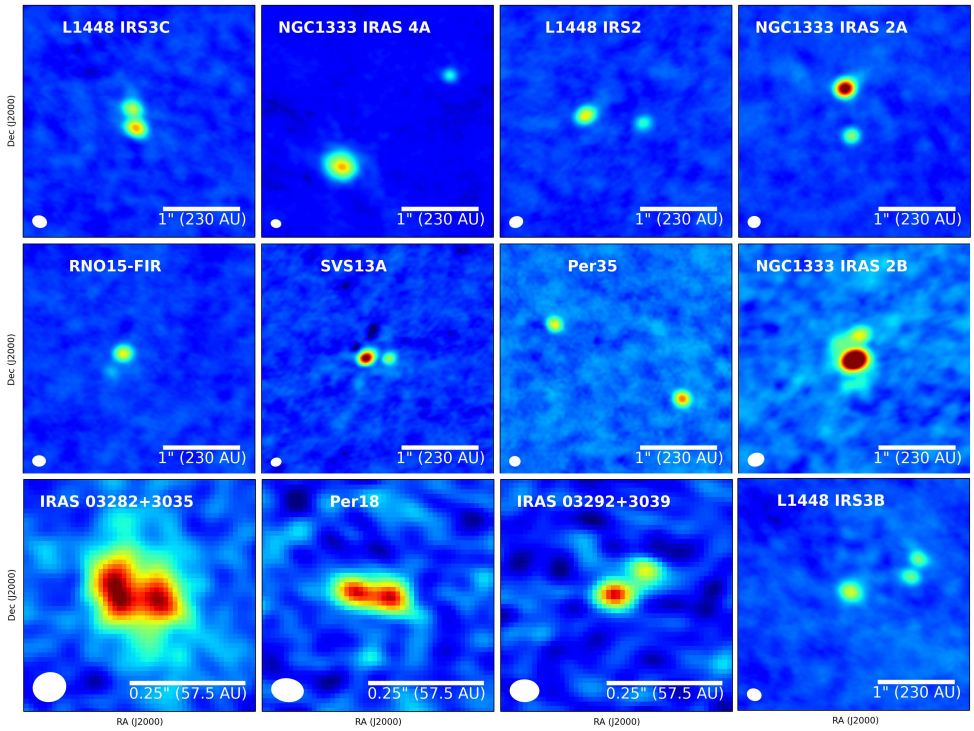

Figure 1: VLA 8 mm images for 12 multiple systems detected by the VANDAM survey in Perseus with separations ranging from 18.5 AU to 420 AU (Tobin et al. 2016). These are a subset of the 18 detected with <500 AU separations. All companion detections are $>5\sigma$; most images are combined A and B configuration images with a beam of $\sim 0.1''$; the images in the bottom row (left three panels) are A configuration-only and have beam sizes of $\sim 0.07''$. More extended ($0.5''$ - $1.5''$) dust emission at 8 mm is detected toward the three sources separated by < 20 AU, but this emission is filtered out in the images with $0.07''$ resolution (first three in bottom row). Only NGC 1333 IRAS 4A and SVS13A were known to be multiples prior to VANDAM. While emission at 8 mm is primarily dust, there is some free-free emission present from the individual protostellar jets. The characterization in Tobin et al. (2016) concludes that it is highly unlikely for any of these multiples to be false positives from extended jet emission, also each resolvable component proposed here is detected in the 1.3 mm continuum (e.g., Figure 2).

3 Constraining the Fragmentation Mechanism

Both kinematic data and dust continuum imaging are necessary to constrain the likely fragmentation processes of a given system. Close proto-binaries that form through disk fragmentation and grow by accreting mass from a circumbinary disk are expected to be embedded within a circumbinary structure that is distinct from the envelope. This disk will be detected in dust continuum, and it should be rotating rapidly, which will be traced by molecular line emission. In addition, since such proto-binaries are formed out a common structure with a single angular momentum direction, their circumstellar disks are expected to be aligned preferentially, which would lead to a preferential alignment of the outflow axes, which we will trace using ^{12}CO. **Detecting large Keplerian circumbinary structures with embedded proto-binaries that drive aligned outflows would favor the disk fragmentation mechanism.** The lack of such structures and/or large outflow misalignment would point to more independent formation, favoring turbulent fragmentation and

subsequent tightening of the orbit through dynamical evolution (Offner et al. 2010). Both mechanisms may be at work. For example, the close binary system L1551IRS5 with a large circumbinary disk and nearly parallel jets may be an example of the former (Rodriguez et al. 1998), and NGC 1333 IRAS2A with nearly orthogonal outflows may be an example of the latter (Tobin et al. 2015). With an unprecedented sample size, our program will determine which mechanism dominates.

3.1 Dust Continuum Structure

The 8 mm data from the VLA are most sensitive to compact sources and we demonstrate in Figure 2 that ALMA 1.3 mm continuum imaging detects significantly more extended dust emission at 1.3 mm for the same resolution. Furthermore, the source L1448 IRS3B shows strong evidence of having a fragmenting circum-triple disk in Figure 2. The proposed ALMA observations will be a factor of $\sim$45 more sensitive to cold dust emission than the VLA observations. Extended dust emission surrounding the multiple systems is already strongly evident in the 25%-completed program. Because all the multiples have separations < 500 AU, the maximum angular scale of $3.5''$ (800 AU) will characterize well the dust emission form the potential circumbinary disk surrounding the protostars and also from most individual components. These observations will provide superior mass estimates for both the circumstellar and circumbinary material compared to the previous VLA observations due to a higher sensitivity to thermal dust at 1.3 mm compared to 8 mm.

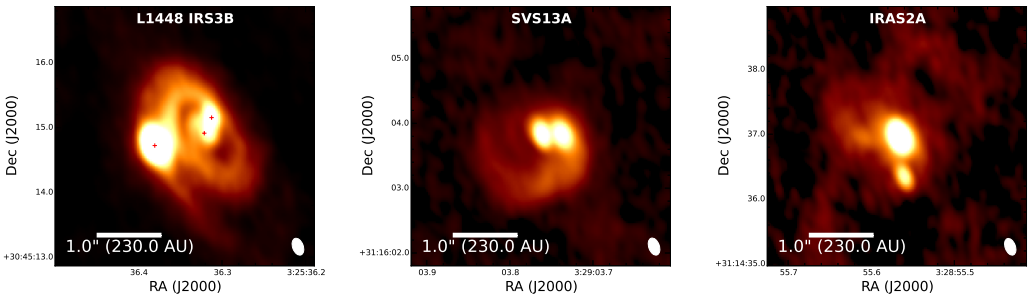

Figure 2: Example 1.3 mm continuum images from our 25%-completed Cycle 2 program (Tobin et al. 2016). Much more extended structures are detected in the continuum with ALMA relative to the VLA; compare to the same sources in Figure 1. The ALMA 1.3 mm continuum for these particular sources shows evidence for complex, asymmetric, and perhaps spiral structure surrounding these multiple systems.

3.2 Disk Kinematics

The kinematics of the surrounding dense gas will be examined with ^{13}CO and C^{18}O ($J = 2 \rightarrow 1$) molecular line emission. We are also targeting three H_2CO ($J = 3 \rightarrow 2$) transitions, SO ($J = 6 \rightarrow 5$), and ^{13}CS ($J = 5 \rightarrow 4$), because these molecules have been found to reside in the inner envelope and disk of L1527 (Sakai et al. 2014), and in our Cycle 2 observations. These higher density tracers will provide a complementary view of the kinematic structure and the three H_2CO can be used to determine the gas temperature when detected.

Gas kinematics are essential to this program. The gas kinematics are needed to determine whether or not circumbinary structures are rotationally-supported, a requirement for disk fragmentation. We will be able to map the molecular line emission from scales as large as 800 AU, encompassing the expected size of the inner envelope, down to the expected scale of the the forming circumbinary and circumstellar disks. Both C^{18}O and ^{13}CO have been used to derive the rotation curves of the recently discovered disks around the Class 0 protostars L1527 IRS and VLA 1623 (Tobin et al. 2012 & Murillo et al. 2013; Ohashi et al. 2014), and disks around Class I protostars (e.g., Harsono et al. 2014). ALMA C^{18}O observations toward the extended circumbinary disk around L1448 IRS3B are shown in Figure 3. A clear velocity gradient is detected across the source and the kinematic

center is on the close (60 AU separation) pair of stars. These data clearly show that the kinematic center of the system is not on the brightest millimeter peak. A possible Keplerian velocity profile is evident in $C^{18}O$ (Figure 3; ^{13}CO as well, but not shown). The rotation profile around the central pair and the apparent circum-triple disk structure is consistent with disk fragmentation. However, the signal-to-noise in our partially completed program (17 mJy/beam per 0.25 km s^{-1} channels) was insufficient to verify Keplerian motion; here we are requesting 2.5× better sensitivity that will will enable us to better constrain the rotation curves. Furthermore, circumbinary Keplerian disks will enable the combined mass of the binary systems to be measured and mass ratios can be estimated from location of the rotation center relative to the companions, assuming circular orbits.

3.2.1 Kinematic Analysis

While Figure 3 shows a PV diagram of $C^{18}O$ that is consistent with Keplerian rotation, the current signal to noise ratio is rather low, a drawback that will be rectified by the higher S/N observations in our proposed program. With higher signal-to-noise, we will quantify the rotation curves in different ways, especially using the method of fitting the centroid of the line emission in each velocity channel and fitting the slopes of velocity versus emission radius to determine the rotation curve and mass (e.g., Ohashi et al. 2014; Murillo et al. 2013). With this method, Keplerian rotation can be distinguished from a rotation profile corresponding to conserved angular momentum, but this method only works for emission >1 km s^{-1} away from line center where the emission is spatially compact.

We will also use analytic thin disk models that include a Keplerian component, with an infalling/angular momentum conserving envelope (e.g., Murillo et al. 2013). We have also developed a framework using the thin disk model to optimally constrain the disk size and protostar mass (Cieza et al. 2016). Also, as we demonstrated for L1527 we will also use molecular line radiative transfer (LIME; Brinch & Hogerheijde 2010) to model more realistic physical structures (Tobin et al. 2012).

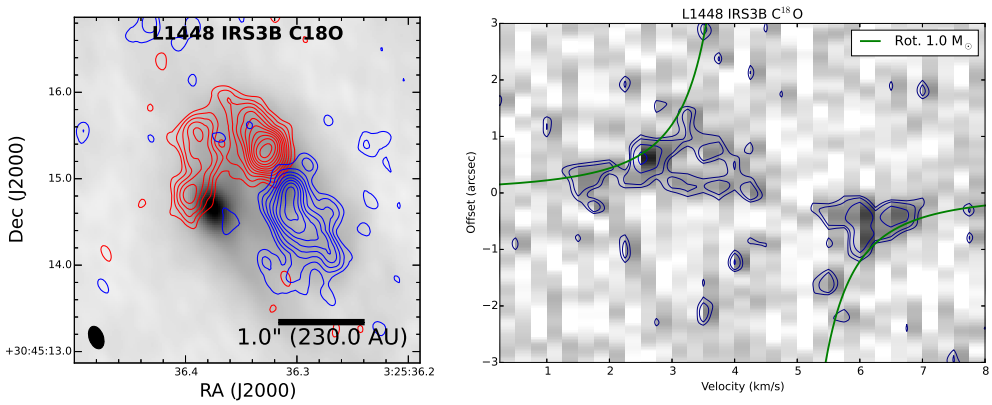

Figure 3: ALMA $C^{18}O$ ($J = 2 \rightarrow 1$) observations toward L1448 IRS3B (Per-emb-33) (Tobin et al. 2016). The left panel shows the integrated intensity of the blue and redshifted $C^{18}O$ emission, plotted with these respective contour colors. The rotation is clearly centered on the close pair detected with the VLA and $C^{18}O$ also traces the extended disk. The right panel shows the position-velocity diagram from the $C^{18}O$ emission. The current data have too low signal to noise to quantitatively fit the mass, but by drawing Keplerian curves we can set an upper limit of ~ 1 $M_\odot$ for the combined mass of the close pair.

References: •André, P. et al. 1993, ApJ, 406, 122 •Brinch, C. & Hogerheijde, M. 2010, A&A, 523, 25 •Chen, X. et al. 2013, ApJ, 768, 110 •Cieza, L. et al. 2016, Nature, 535, 258 •Kratter, K. M. et al. 2010, ApJ, 708, 1585 •Lada, C. J. 2006, ApJ, 640, 63 •Looney et al. 2000, ApJ, 529, 477 •Maury, A. J. et al. 2010, A&A, 512, 40 •Murillo, N. et al. 2013, A&A, in press. •Offner, S. et al. 2010, ApJ, 725, 1485 •Raghavan, D, et al. 2010, ApJS, 190, 1 •Tobin, J. J. et al 2012, Nature, 492, 83 •Tobin, J. J. et al 2015, ApJ, 798, 61 •Tobin, J. J. et al 2016, ApJ, 818, 73 •Tobin, J. J. et al 2016, Nature, 538, 483 •Tohline, et al. 2002, ARA&A, 40, 349 •Vorobyov, E. 2010, ApJ, 723, 1294

2017.1.00053.S

SG : 1 of 1 Perseus Binaries Band 6

Observe dust continuum and molecular lines toward binary protostars in the Perseus molecular cloud.

Science Goal Parameters

Ang.Res.	LAS	Requested RMS	RMS Bandwidth	Rep.Freq.	Cont. RMS	Cont. Bandwidth	Poln.Prod.	Non-standard mode
0.2500"	3.5"	6.9 mJy, 2.8 K	250 m/s, 183.1 kHz	219.560358 GHz	53.343 µJy, 21.6 mK	2.695 GHz	XX,YY	No

Use of 12m Array (43 antennas)

t_total(all configs)	t_science(C43-5)	t_total()	Imaged area	#12m pointing	12m Mosaic spacing	HPBW	t_per_point	Data Vol	Avg. Data Rate
5.9 h	3.5 h	0.0 h	8.8 "	12	offset	26.5 "	1209.1 s	389.7 GB	18.8 MB/s

Use of ACA 7m Array (10 antennas) and TP Array

t_total(ACA)	t_total(7m)	t_total(TP)	Imaged area	#7m pointing	7m Mosaic spacing	HPBW	t_per_point	Data Vol	Avg. Data Rate

Spectral Setup : Spectral Line

BB	Center Freq Rest GHz	spw name	Eff #Ch p.p.	Bandwidth	Resolution	Vel. Bandwidth	Vel. Res.	Res. El. per FWHM
1	220.398684	13CO v=0 2-1	1920	58.59 MHz	61.035 kHz	79.7 km/s	0.083 km/s	48
1	219.560358	C18O 2-1	1920	58.59 MHz	61.035 kHz	80.0 km/s	0.083 km/s	48
2	219.949442	SO 3Σ v=0 6(5)-5(4)	960	58.59 MHz	0.122 MHz	79.9 km/s	0.166 km/s	24
2	218.760066	H2CO 3(2,1)-2(2,0)	960	58.59 MHz	0.122 MHz	80.3 km/s	0.167 km/s	24
2	218.475632	H2CO 3(2,2)-2(2,1)	960	58.59 MHz	0.122 MHz	80.4 km/s	0.168 km/s	24
2	218.222192	H2CO 3(0,3)-2(0,2)	960	58.59 MHz	0.122 MHz	80.5 km/s	0.168 km/s	24
3	231.250000	13CS v=0 5-4	1920	234.38 MHz	0.282 MHz	303.9 km/s	0.366 km/s	11
3	230.556000	CO v=0 2-1	1920	234.38 MHz	0.282 MHz	304.8 km/s	0.367 km/s	11
4	233.500000	cont	128	1875.00 MHz	31.250 MHz	2407.4 km/s	40.124 km/s	0

12 Targets

No.	Target	Ra,Dec (ICRS)	V,def,frame --OR--z
1	1-Per12_NGC_13...	03:29:10, 31:13:30	6.00 km/s,lsrk,RADIO
2	2-Per17	03:27:39, 30:13:03	6.00 km/s,lsrk,RADIO
3	3-Per18	03:29:11, 31:18:31	6.00 km/s,lsrk,RADIO
4	4-Per22_L1448_I...	03:25:22, 30:45:13	6.00 km/s,lsrk,RADIO
5	5-Per27_NGC133...	03:28:55, 31:14:37	6.00 km/s,lsrk,RADIO
6	6-Per33_L1448_I...	03:25:36, 30:45:14	6.00 km/s,lsrk,RADIO
7	7-Per35	03:28:37, 31:13:30	6.00 km/s,lsrk,RADIO
8	8-Per36_NGC_13...	03:28:57, 31:14:15	6.00 km/s,lsrk,RADIO
9	9-Per44_SVS_13A	03:29:03, 31:16:03	6.00 km/s,lsrk,RADIO
10	10-Per107_L1448...	03:25:35, 30:45:34	6.00 km/s,lsrk,RADIO
11	11-Per2_IRAS_03...	03:32:17, 30:49:47	5.00 km/s,lsrk,RADIO
12	12-Per5_IRAS032...	03:31:20, 30:45:30	8.00 km/s,lsrk,RADIO

Expected Source Properties

	Peak Flux	SNR	Pol.	Pol. SNR	Linewidth	RMS (over 1/3 linewidth)	linewidth / bandwidth used for sensitivity
Line	21.00 mJy	7.5	0.0%	0.0	4 km/s	2.8 mJy, 1.1 K	16.00
Continuum	750.00 uJy	14.1	0.0%	0.0			

Dynamic range (cont flux/line rms): 0.2

1 Tuning

Tuning	Target	Rep. Freq. Sky GHz	RMS (Rep. Freq.)	RMS Achieved
1	1,2,3,4,5,6,7,...	219.555964	6.47 mJy, 2.6 K	6.46 mJy - 8.18 mJy

Justification for requested RMS and resulting S/N (and for spectral lines the bandwidth selected) for the sen...

Our overall time estimate is driven by the sensitivity requirement of the C18O (J=2-1) observations. Our 25%-completed Cycle 2 program yielded a sensitivity of 17 mJy/beam in a 0.25 km/s channel. For the proposed sources, 2.5x better sensitivity, will enable much better detections of the C18O emission that is currently present at ~3 sigma level. Thus we require a sensitivity of 6.9 mJy/beam in 0.25 km/s channels. This sensitivity will enable us to detect the high-velocity line wings of C18O and 13CO associated with disk rotation out to at least +/- 2 km/s from line center (see Figure 3) and this sensitivity is comparable to what has enabled masses to be measured for other protostars with ALMA (e.g., VLA 1623 and L1527) and our sources have similar luminosities to these systems. A signal-to-noise ratio >= 5 is necessary to centroid the position of line emission at the high-velocities to an accuracy of ~13 AU; the relative positional accuracy is ~ beam/signal-to-noise (Reid et al. 1988).

The integration time necessary for the line observations will yield a continuum sensitivity of 0.06 mJy with the 2 GHz of bandwidth. This sensitivity is more than enough to detect the individual sources, in addition to an compact circumbinary disk structures (see Figure 2).

Justification of the chosen angular resolution and largest angular scale for the source(s) in this Science Goal.

Our observations will be conducted at 0.25" resolution in order to resolve the individual proto-binary sources and map kinematics in the protostellar disks. The declination of Perseus is about 31 degrees (max. elevation 36 degrees), casuing the beam to be elongated in the north-south direction yielding 0.35" X 0.2" resolution; this elongation will not affect our ability to resolve the close binary sources. The largest angular scale in the observations will be 3.5", encompassing the entire area shown in Figure 1 subpanels. This is more than enough spatial sensitivity to detect circumbinary structure on the scale of ~800 AU around these sources. The surrounding envelopes will be resolved-out, but this will not affect our science goal of examining the small scale structure directly associated with the protobinary sources. The C18O and 13CO emission that traces disk rotation will be spatially unresolved in the line wings at velocities greater than +/- 0.5 km/s from line center.

Justification of the correlator set-up with particular reference to the number of spectral resolution element...

We will observe the C18O and 13CO (J=2-1) lines the same base-band, allocating half the correlator resources to each line and observing with a 60 MHz bandwidth, 0.083 km/s velocity resolution. We are also observing the SO (6-5) line, and three H2CO (3-2) lines with 0.168 km/s resolution in the second baseband. The 13CS (5-4) and 12CO (2-1) will be observed in the third baseband with 0.091 km/s and 0.158 km/s resolution, respectively. Depending on the brighness of a particular source, we plan to bin the data to 0.25 km/s in post-processing, this corresponds to approximately 3 channels for C18O and 13CO. The additional lines sampled at 0.168 km/s may or may not be averaged depending on source strength.

The lines execpt 12CO, which traces outflows, are expected to trace gas kinematics on the scale of disks toward the protostellar sources. The line widths are expected to be +/-2 km/s owing to rotation of the protostellar disks; the high spectral resolution is necessary because the position of the line emission changes with velocity and mapping the position-velocity change is crucial to determining if the sources have rotationally-supported circumbinary disks. The 12CO line is expected to trace the outflow emission and will be spread over a larger velocity range (> +/- 10 km/s) necessitating lower velocity resolution and broader overall bandwidth.

We are allocating one baseband to observe the 1.3 mm continuum emission with a total of 2 GHz of bandwidth in order to examine the compact dust emission surrounding the protostars.